Current System Analysis

FOR POWER-WORLD GYM

IM/2022/001 - A.M.K.C.S.B.Abeykoon

Introduction

This report documents the current system workflows of Power World Gym – Kiribathgoda through use-case and activity diagrams. The analysis focuses on manual processes for member registration, attendance tracking, workout management, and instructor coordination. The goal is to identify inefficiencies to inform the proposed Gym Management Suite. The actors for the diagrams are defined as **Member**, **Instructor**, **Admin**, and **Top Management**.

Current System - Overview

Figure 1 - QR code which redirects to the payment system

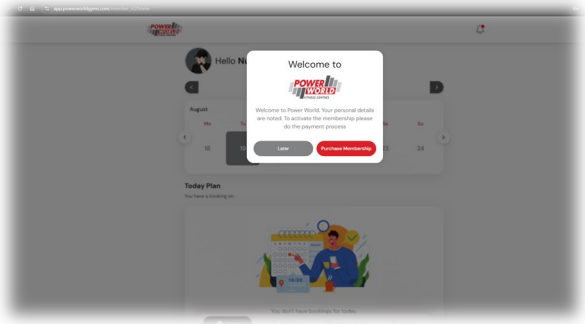


Figure 2 - Current system homepage – handling only subscriptions

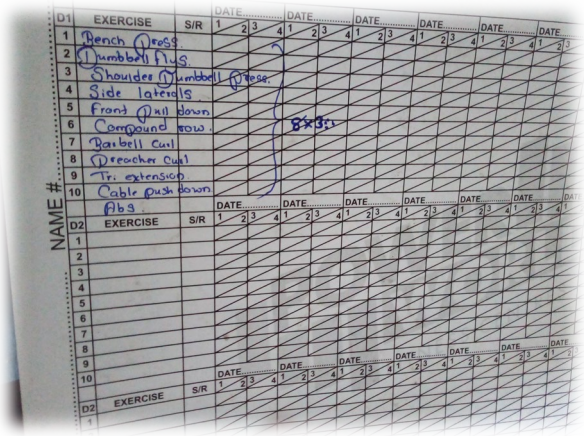


Figure 3 - Traditional routine schedule

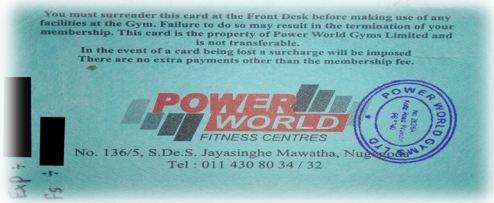


Figure 4 – Physical registration card

Use-case diagram

Documentation

Actors:

- **Member**: Gym user.
- **Instructor**: Staff handling registrations, routines, and equipment.
- **Admin**: Manages records (limited role in current system).
- **Top Management**: Oversees operations (minimal direct involvement).

Use Case 1: On-Site Member Registration

- **Goal**: Register a new member manually.
- **Preconditions**: Member physically present; instructor available.
- **Postconditions**: Member details recorded; subscription card issued.
- **Actors**: Member, Instructor.
- **Relationships**: None.

Use Case 2: Record Attendance (Optional)

- **Goal**: Log entry in attendance book.
- **Preconditions**: Member enters gym; book is accessible.
- **Postconditions**: Attendance optionally recorded.
- **Actors**: Member.
- **Relationships**: None.

Use Case 3: Provide Workout Schedule

- **Goal**: Issue a generic workout card.

- **Preconditions**: Member requests schedule; instructor available.
- **Postconditions**: Card given (rarely updated).
- **Actors**: Member, Instructor.
- **Relationships**: None.

Use Case 4: Ad-Hoc Routine Modification

- **Goal**: Member changes workout without monitoring.
- **Preconditions**: Member has schedule card.
- **Postconditions**: Unregulated routine adjustments.
- **Actors**: Member, Instructor (sometimes)
- **Relationships**: Extends *Provide Workout Schedule*.

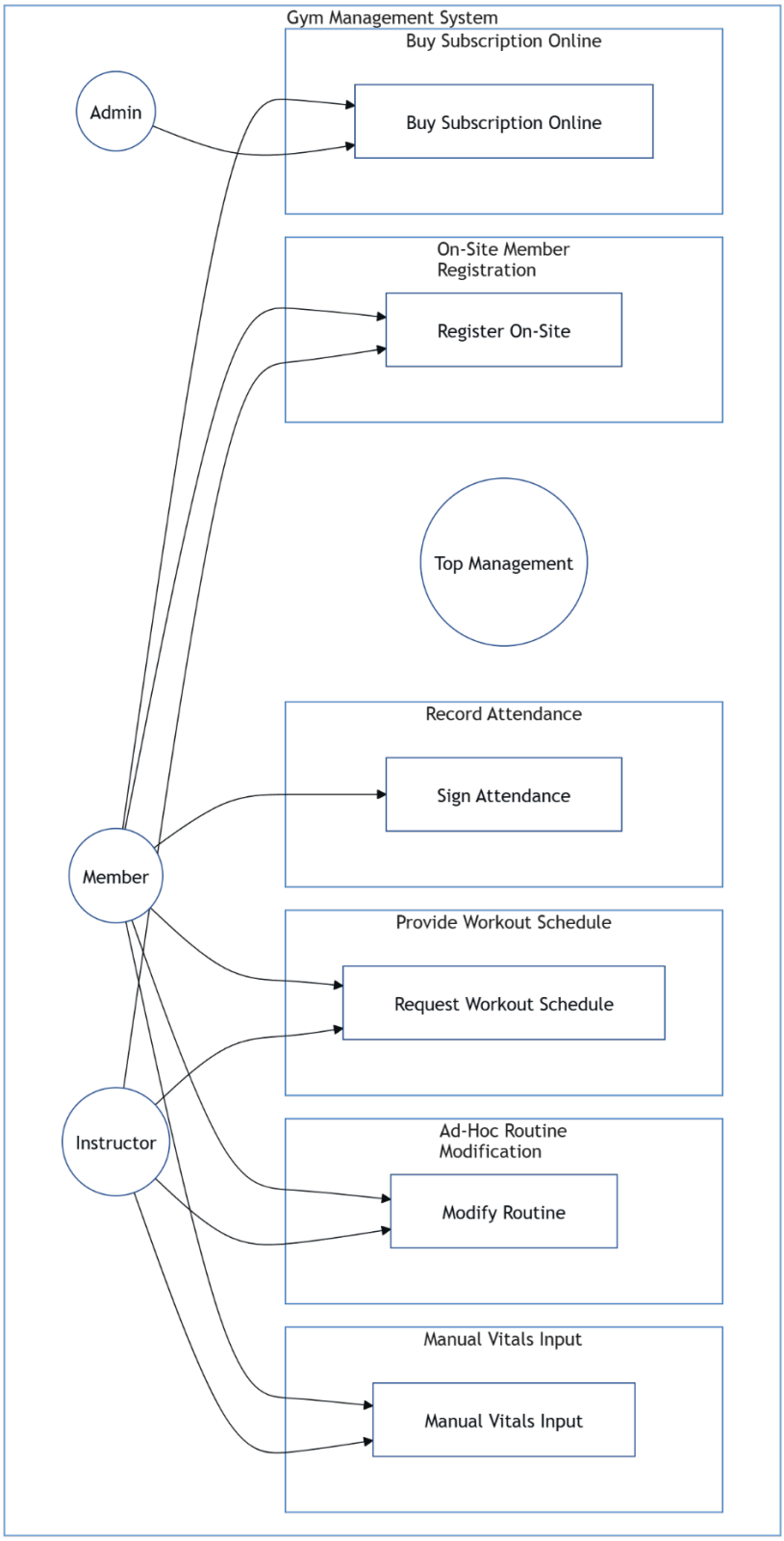
Use Case 5: Manual Vitals Input (Planned)

- **Goal**: Record BMI/metrics (system not implemented).
- **Preconditions**: N/A (process incomplete).
- **Postconditions**: N/A.
- **Actors**: Member, Instructor.
- **Relationships**: None.

Use Case 6: Buy Subscription Online

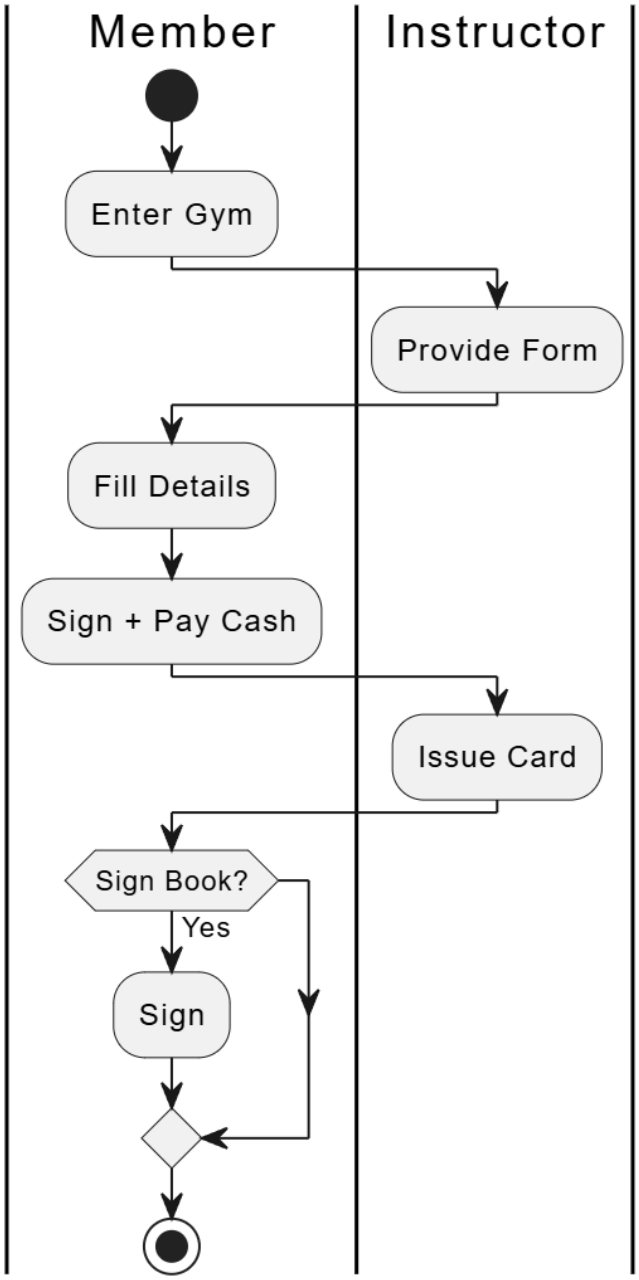
- **Goal**: Purchase gym membership digitally.
- **Preconditions**: Internet access; valid payment method.
- **Postconditions**: Subscription recorded; no physical verification at gym.
- **Actors**: Member, Admin
- **Relationships**: None.

The use-case diagram

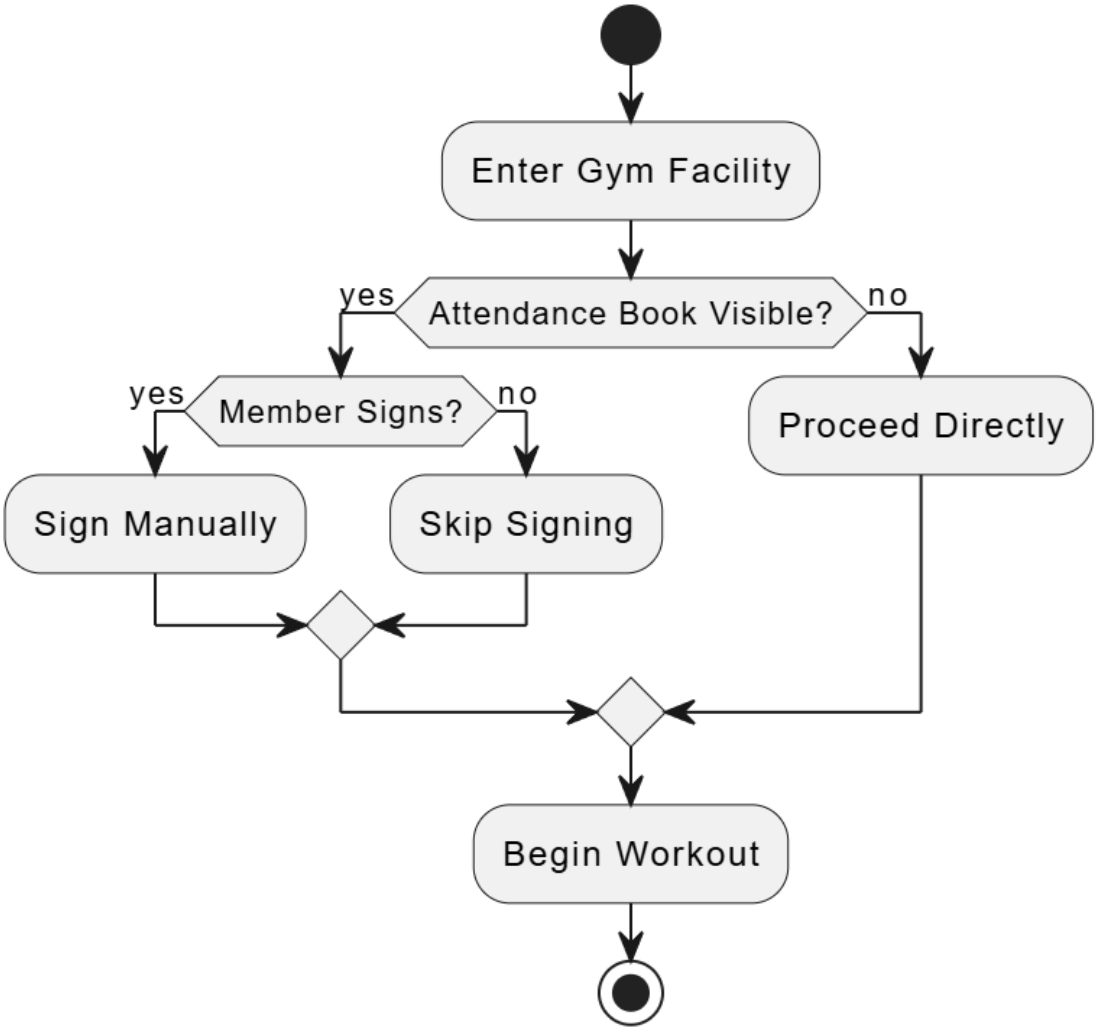


Activity diagrams

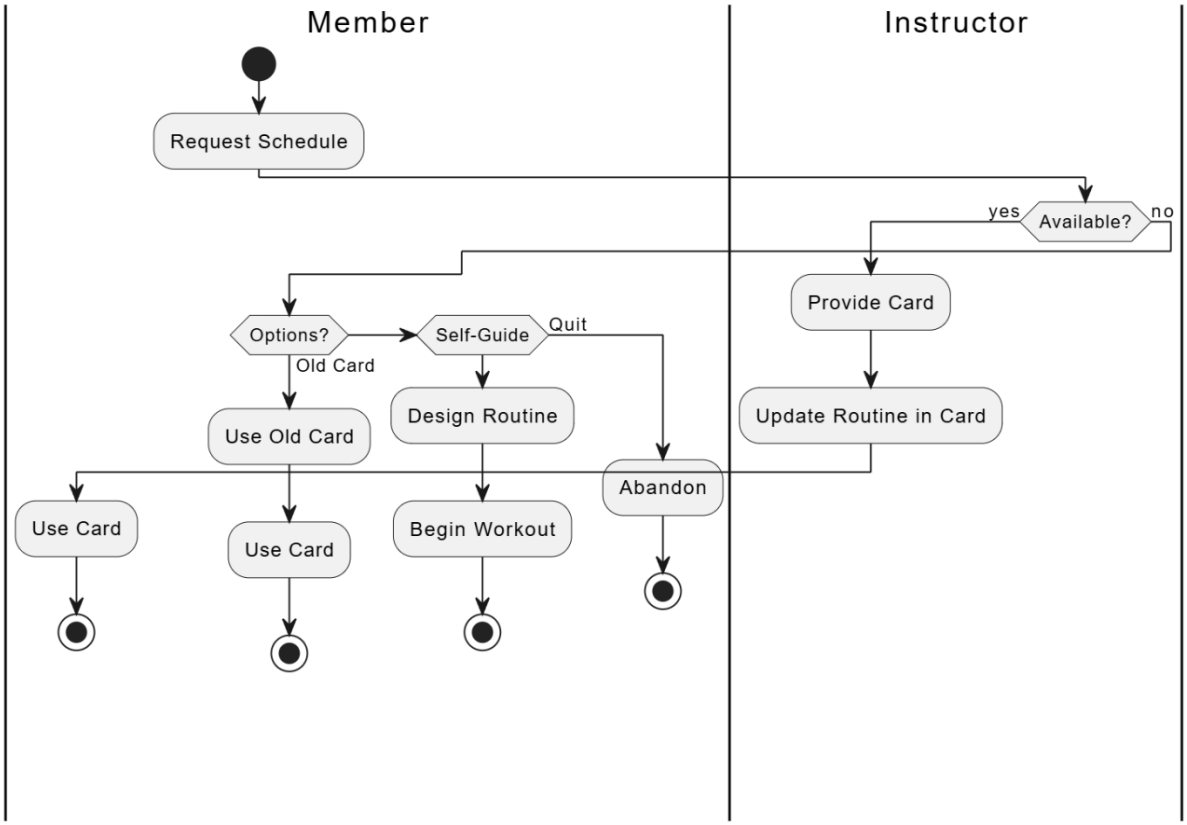
Use Case 1: “On-Site Member Registration” Activity Diagram



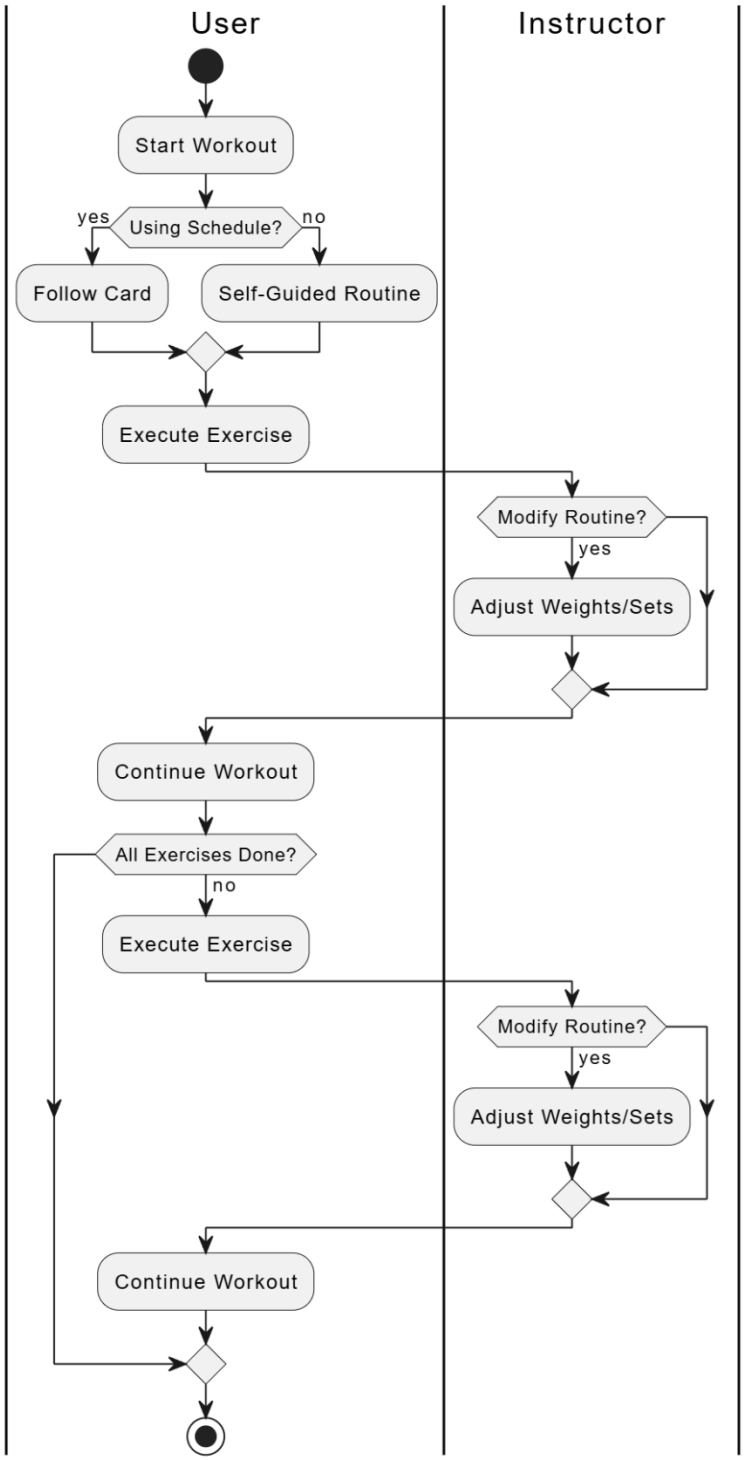
Use Case 2: “Record Attendance (Optional)” Activity Diagram



Use Case 3: “Provide Workout Schedule” Activity Diagram

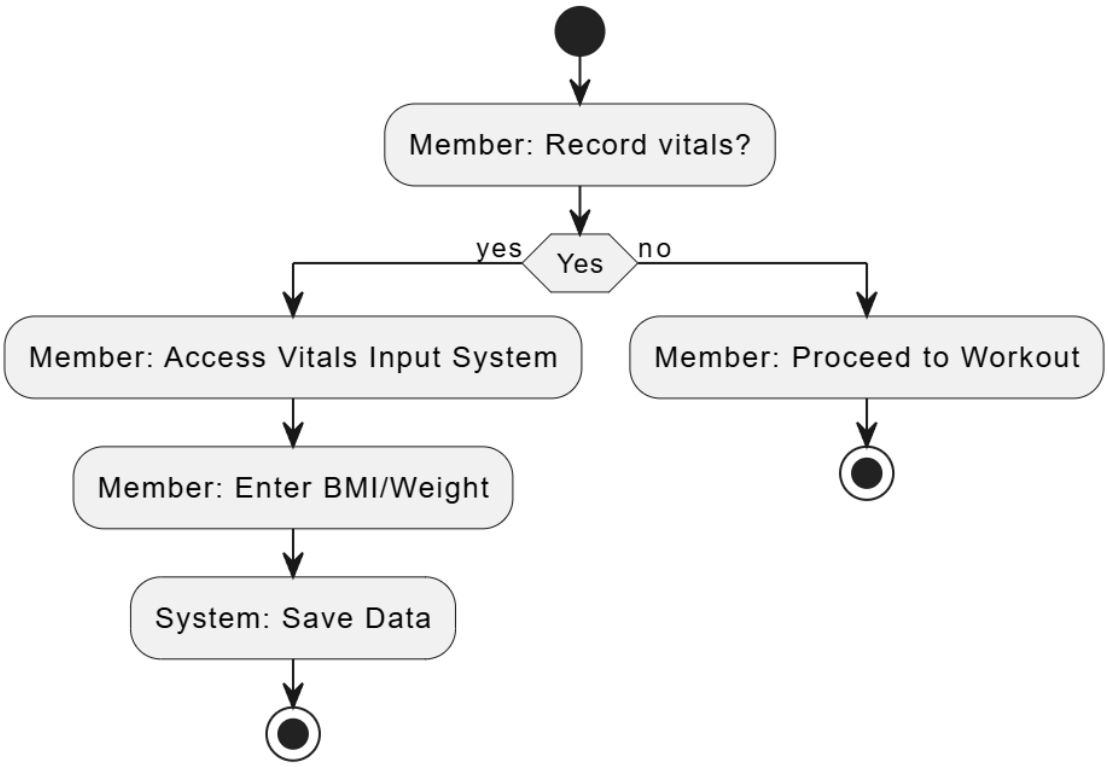


Use Case 4: “Ad-Hoc Routine Modification” Activity Diagram

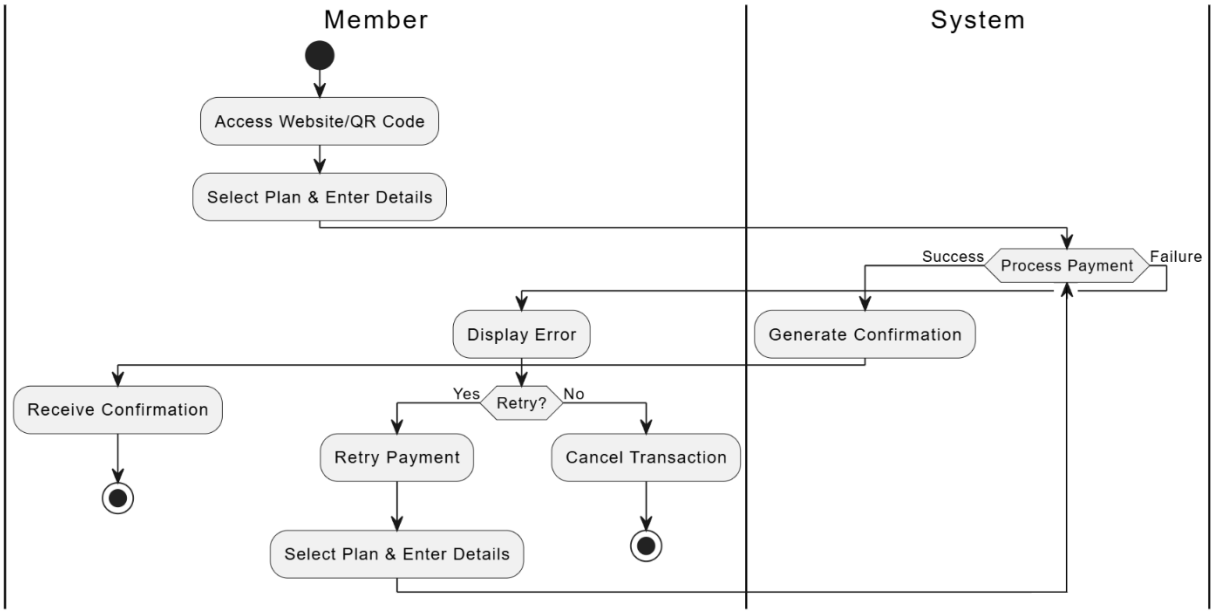


(assuming instructor is available)

Use Case 5: “Manual Vitals Input (Planned)” Activity Diagram



Use Case 6: “Buy Subscription Online” Activity Diagram



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